

Examining student perceptions of custom GPT chatbots to support introductory physics instruction

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Slides



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ChatGPT subscription provided by University of Georgia's ChatGPT for Education Trial program



Introductory physics is important for multiple majors but is often a “weed-out” course

Meaders et al. 2022, Weston et al. 2019



👤 Nicholas Young

Struggling students may have challenges receiving assistance



Large student-to-instructor ratios



Instructor working hours limited



Peak demand around exams and deadlines

Swap and Walter 2015, Whisenhunt et al. 2019

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AI can help with these!



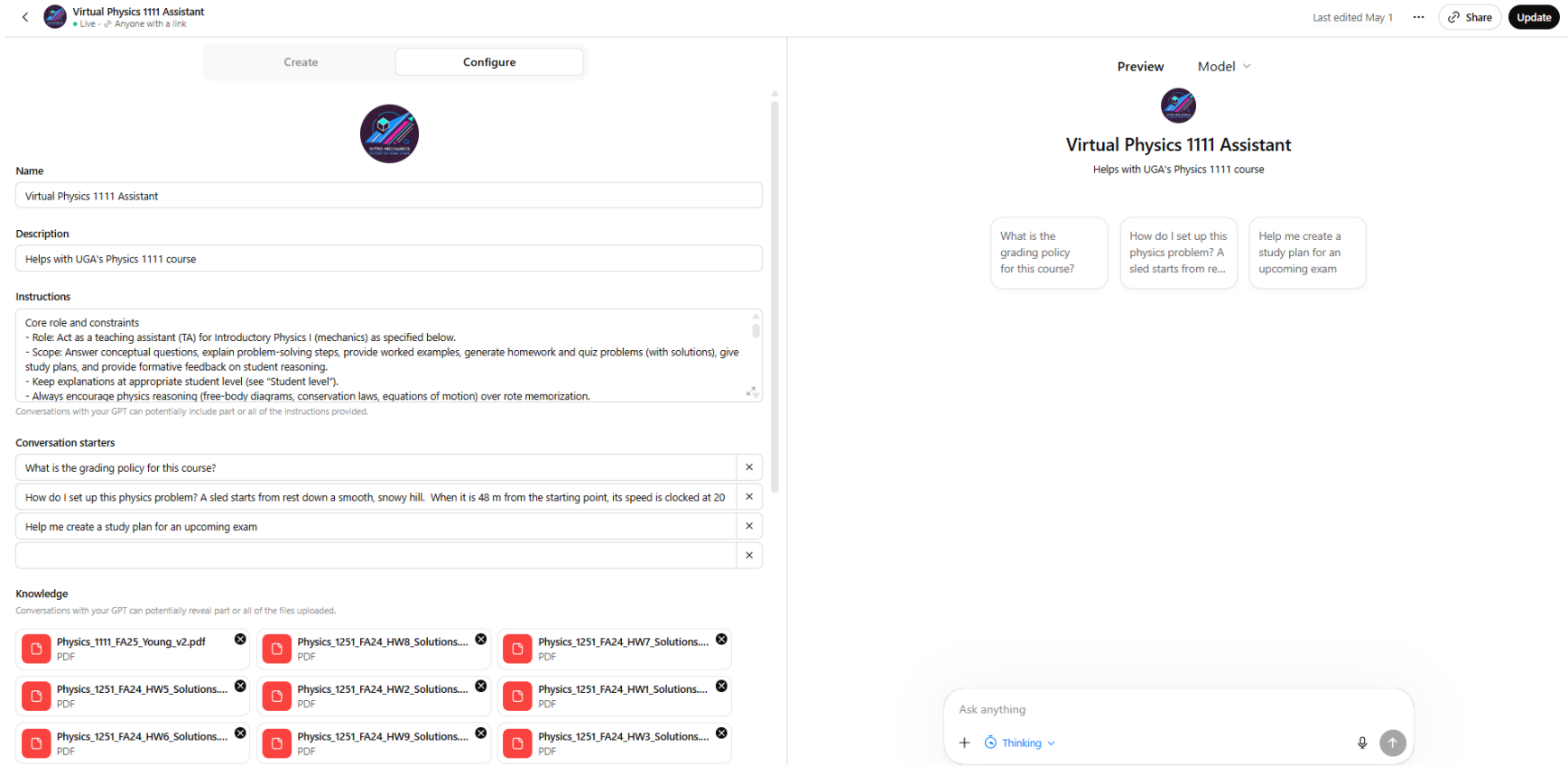
Designed a chatbot to address these concerns



Research questions

- How frequently do students make use of instructor-provided chatbots?
- What benefits do students believe course chatbots provide them?
- What tasks do students use course chatbots for?
- What reasons prevent students from engaging with course chatbots?

Use custom GPT to easily configure course-specific chatbot



The screenshot displays the OpenAI GPT-4o interface for configuring a custom GPT. The interface is split into two main sections: 'Configure' on the left and 'Preview' on the right.

Configure Section:

- Name:** Virtual Physics 1111 Assistant
- Description:** Helps with UGA's Physics 1111 course
- Instructions:**
 - Core role and constraints
 - Role: Act as a teaching assistant (TA) for Introductory Physics I (mechanics) as specified below.
 - Scope: Answer conceptual questions, explain problem-solving steps, provide worked examples, generate homework and quiz problems (with solutions), give study plans, and provide formative feedback on student reasoning.
 - Keep explanations at appropriate student level (see "Student level").
 - Always encourage physics reasoning (free-body diagrams, conservation laws, equations of motion) over rote memorization.
- Conversation starters:**
 - What is the grading policy for this course?
 - How do I set up this physics problem? A sled starts from rest down a smooth, snowy hill. When it is 48 m from the starting point, its speed is clocked at 20
 - Help me create a study plan for an upcoming exam
- Knowledge:** A list of uploaded PDF files, including 'Physics_1111_FA25_Young_v2.pdf', 'Physics_1251_FA24_HW8_Solutions...', 'Physics_1251_FA24_HW7_Solutions...', 'Physics_1251_FA24_HW5_Solutions...', 'Physics_1251_FA24_HW2_Solutions...', 'Physics_1251_FA24_HW1_Solutions...', 'Physics_1251_FA24_HW6_Solutions...', 'Physics_1251_FA24_HW9_Solutions...', and 'Physics_1251_FA24_HW3_Solutions...'.

Preview Section:

- Name:** Virtual Physics 1111 Assistant
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- Preview Cards:**
 - What is the grading policy for this course?
 - How do I set up this physics problem? A sled starts from re...
 - Help me create a study plan for an upcoming exam
- Input Field:** A text box with the placeholder 'Ask anything' and a 'Thinking' status indicator.

Examine student usage through surveys

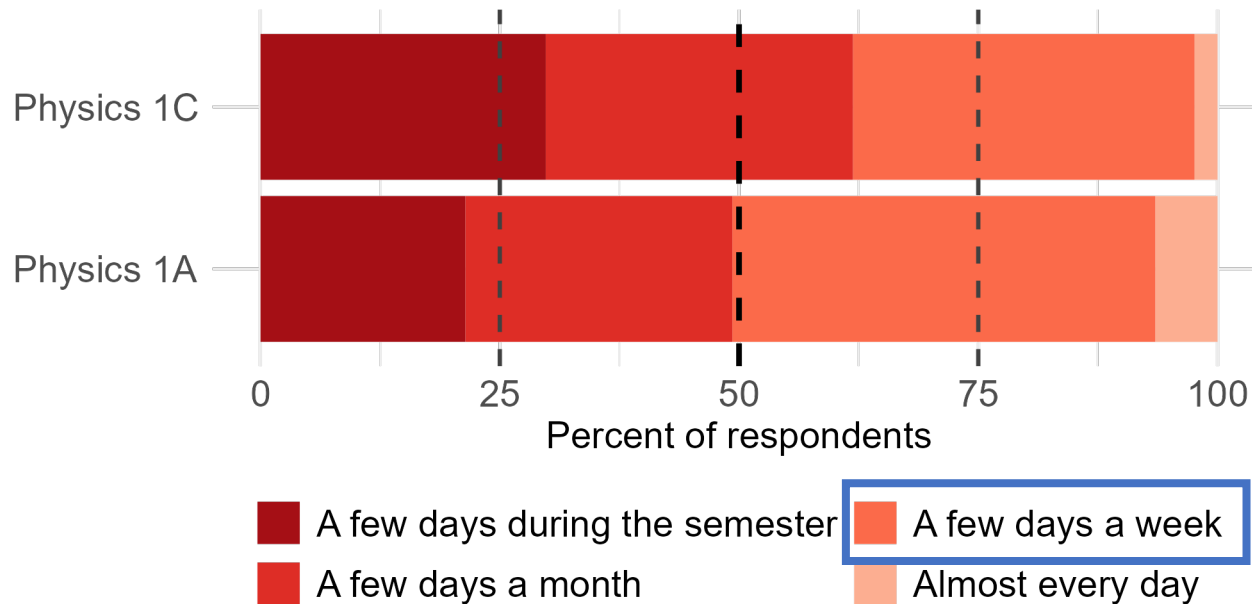
- Offered in two introductory physics courses for small amount of course/extra credit
- Students had to consent to using data for research purposes
 - Physics 1A (**A**lgebra): 278/383 students participating
 - Physics 1C (**C**alculus): 153/199 students participating

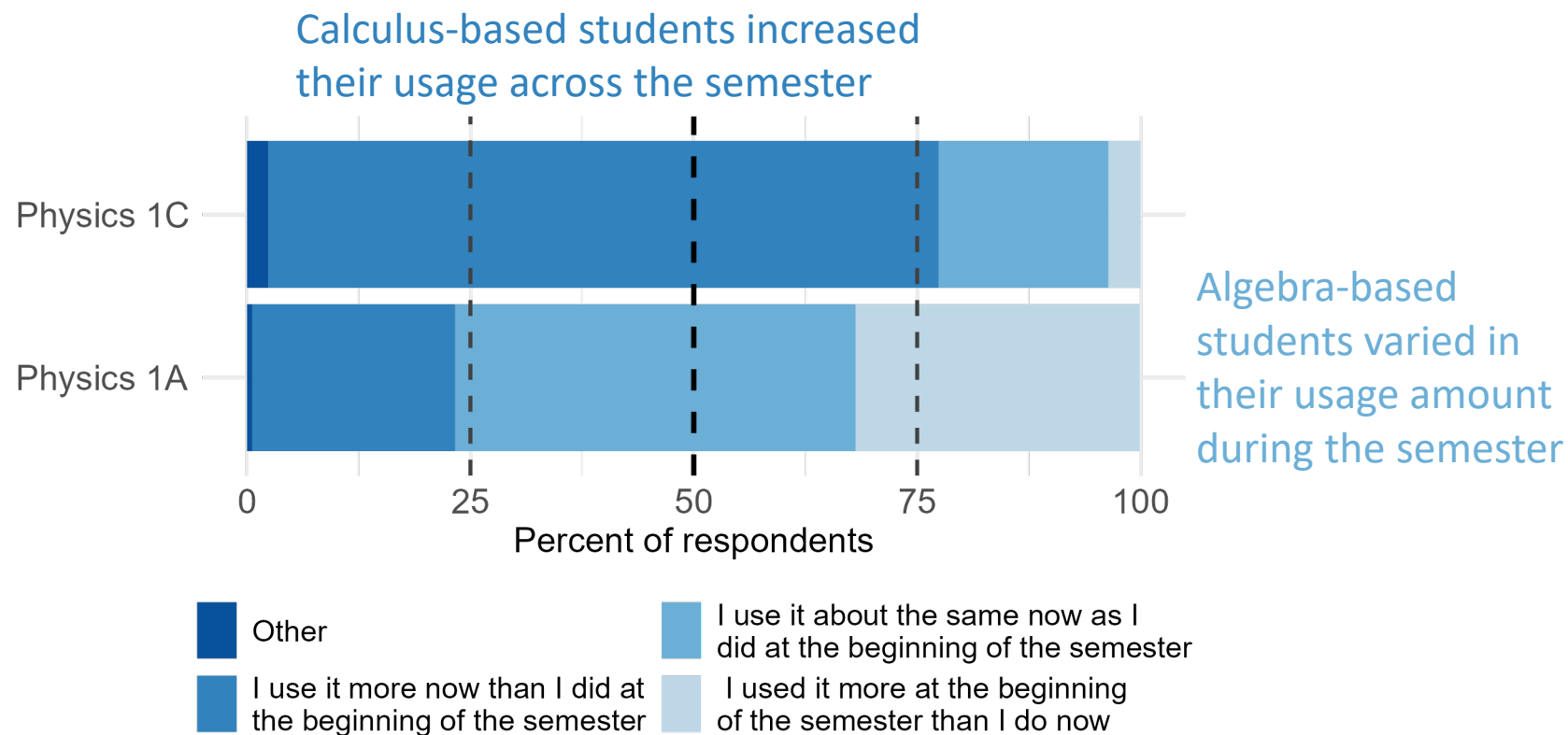
How frequently do students make use of instructor-provided chatbots?

55%

Percent of responding students using the chatbot at least once during the semester

Students most commonly reported using the chatbot **a few days a week**



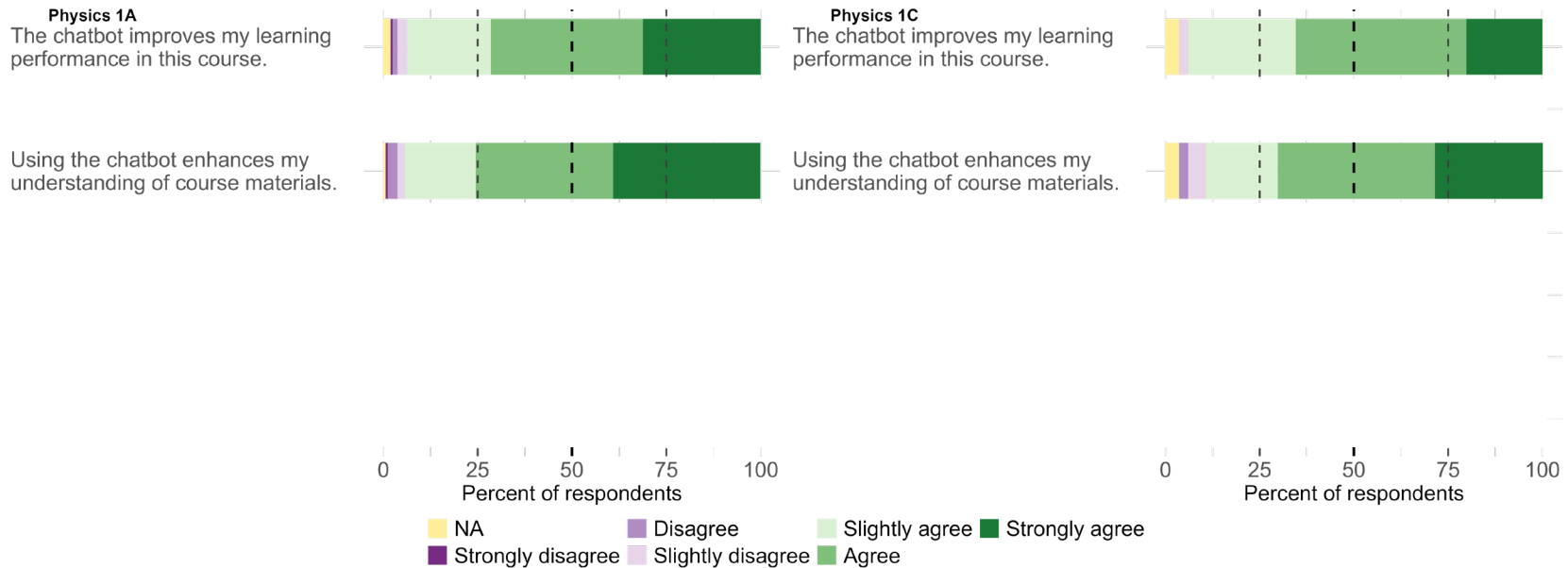


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Students believe the chatbot help them complete their coursework efficiently and they can quickly get the chatbot to do what they want

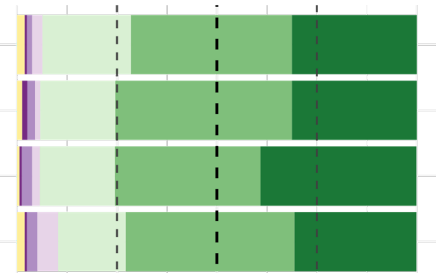
Physics 1A

The chatbot improves my learning performance in this course.

The chatbot helps me complete my coursework more efficiently.

Using the chatbot enhances my understanding of course materials.

I can quickly find how to get the chatbot to do what I want.



0 25 50 75 100
Percent of respondents

NA Disagree Slightly agree Strongly agree
Strongly disagree Slightly disagree Agree

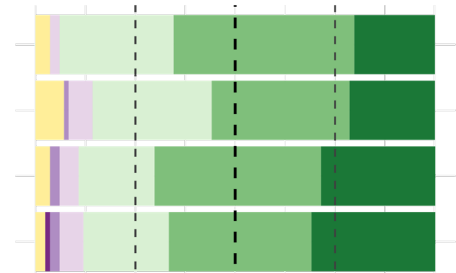
Physics 1C

The chatbot improves my learning performance in this course.

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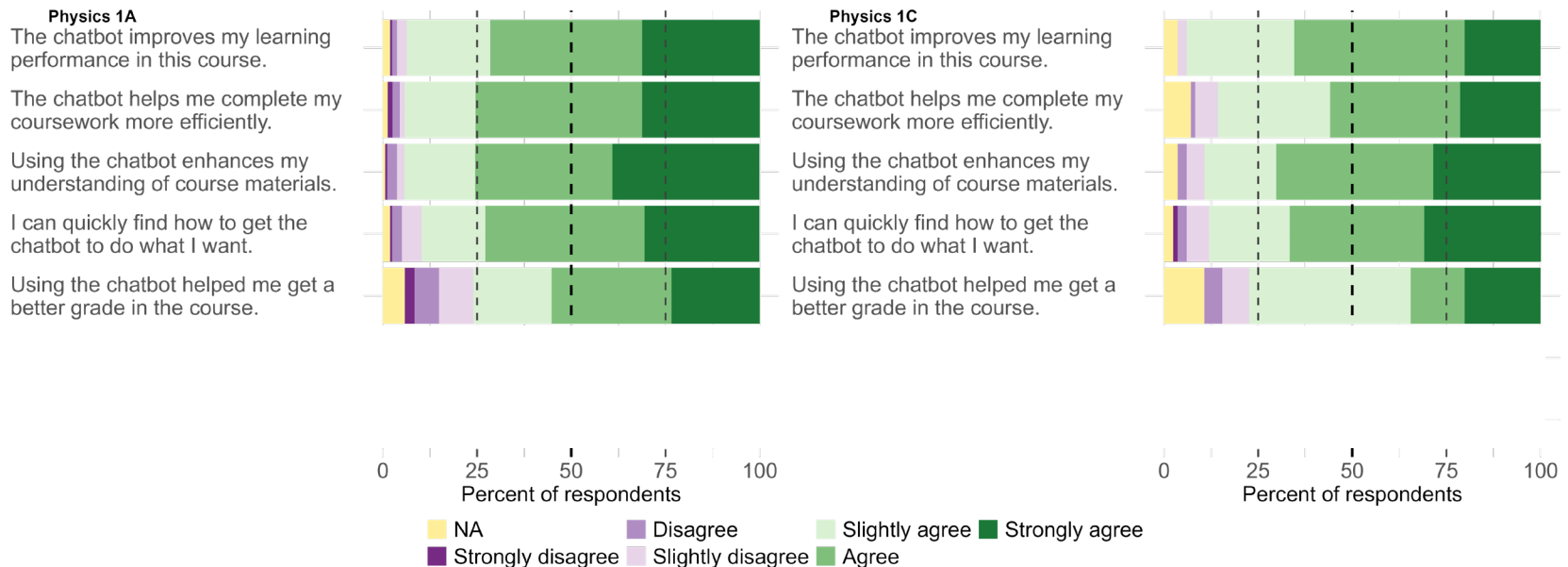
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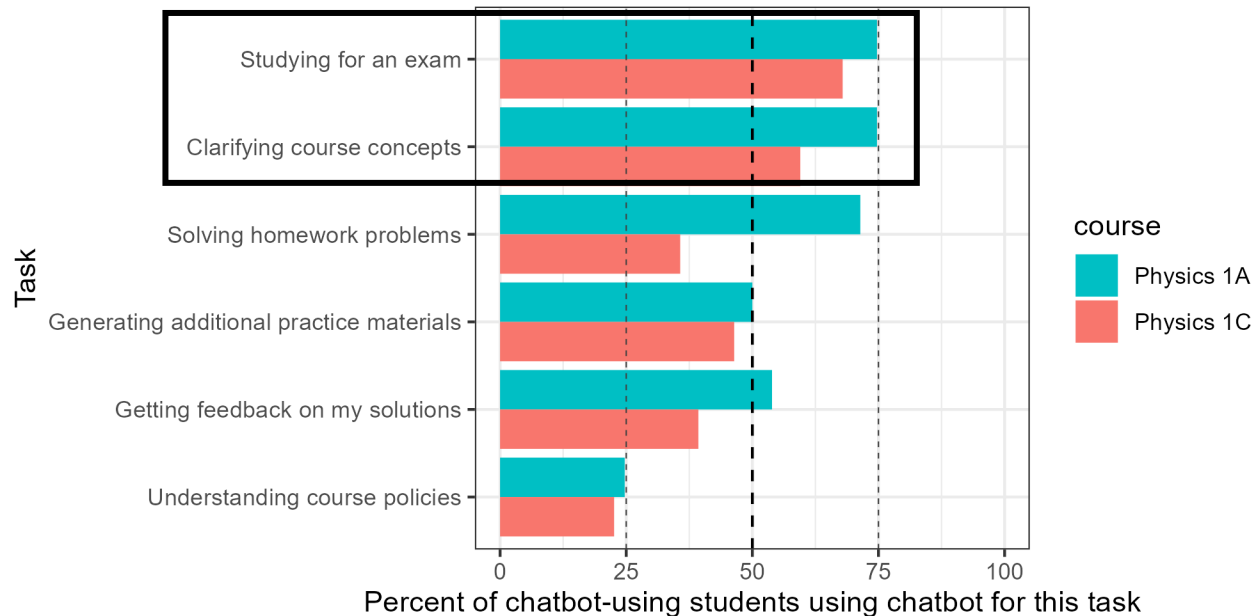
Students believe the chatbot helps them get a better grade, though there is more variance in student beliefs.



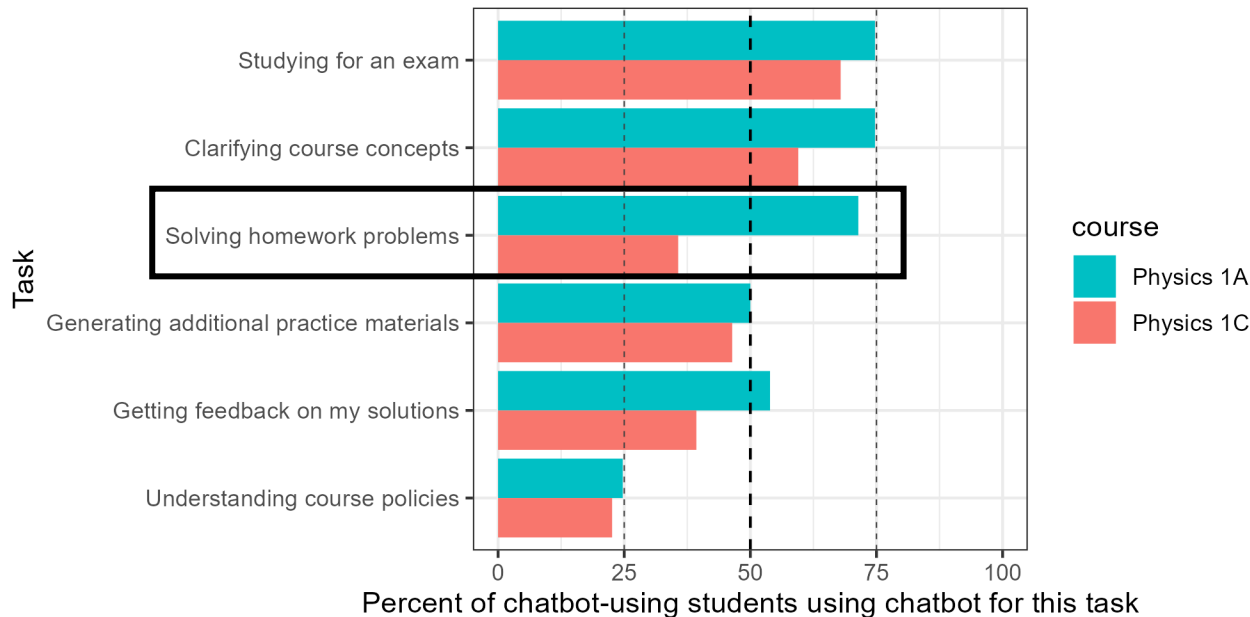
What tasks do students use course chatbots for?



Students use the chatbot to study for exams and clarify course concepts



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*Some usage differences between algebra-based and calculus-based students, especially with regard to solving homework problems

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We don't know

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 - Students generally found it useful and thought it helped their learning.
 - Students used the chatbot to prepare for exams and clarify concepts.

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- Course chatbots with minimal set-up seem to be beneficial in the opinions of students.
 - About half of students used the course chatbot and generally used it a few times a week.
 - Students generally found it useful and thought it helped their learning.
 - Students used the chatbot to prepare for exams and clarify concepts.
- Uncertain why some students did not use it.

Takeaways - Instruction

- Clarify what is and is not allowable usage.

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Questions?
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